Passion Project

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1 Introduction

This is a passion project of mine, where I go out of my way to educate myself. I have been reading about numerical differentiation, approximation etc. Based on the things, that I have read, I am making programs with no real application, other than to get acquainted with numerical methods. My goal is that this knowledge will come in help later down the line in my educational/professional life. With that in mind this document will contain information about the different programs, that I have made, and maybe some comments about the work, put in them.
Contains:

- Euler Approximation Matlab
- 1-D Simplified Heat Equation Matlab

2 Euler Approximation Method

This method is used to numerically solve ordinary differential equations of this type: $y' = f(x, y)$. Using the following property of the derivative we can approximate the original function for very small intervals $x \to x_0$:

$$y'(x) = lim_{x \to x_0} \frac{y(x) - y(x_0)}{x - x_0} \tag{1}$$

$$\Rightarrow y(x) \approx y'(x) * (x - x_0) + y(x_0) \tag{2}$$

$$\Rightarrow y(x) \approx f(x, y) * (x - x_0) + y(x_0) \tag{3}$$

Therefore if we use the given to us function $f(x, y)$ and the initial condition for example $y(0) = y_0$, we can evaluate the next value $y_1, y_2, ...$ etc. Let us take a look at the following example:

$$y' - y = -\frac{1}{2}e^{x/2}sin(5x) + 5e^{x/2}cos(5x) \tag{4}$$

$$\Rightarrow f(x, y) = y - \frac{1}{2}e^{x/2}sin(5x) + 5e^{x/2}cos(5x) \tag{5}$$

$$\Rightarrow y_{n+1} = y_n + f(t_n, y_n)(t_{n+1} - t_n); \tag{6}$$

When put into a Matlab script, we will make the following boundaries, in order to form a visualization:

- The initial given value is $y(1) = 0$.
- The graph takes place in $x \in [0, 10]$.
- The difference between x_n and x_{n+1} is n.

When we take n for 1000, we get the following the following difference of the least squares and the following graph:

$$\chi_1^2 = -10.1399 \tag{7}$$
$$\chi_2^2 = 0.8782 \tag{8}$$
$$\chi_3^2 = 0.9988 \tag{9}$$

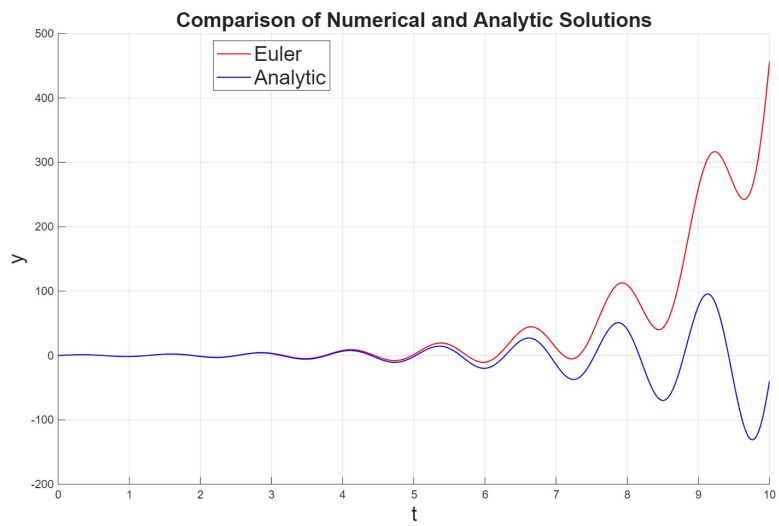

Figure 1: Euler Method, Iteration 1

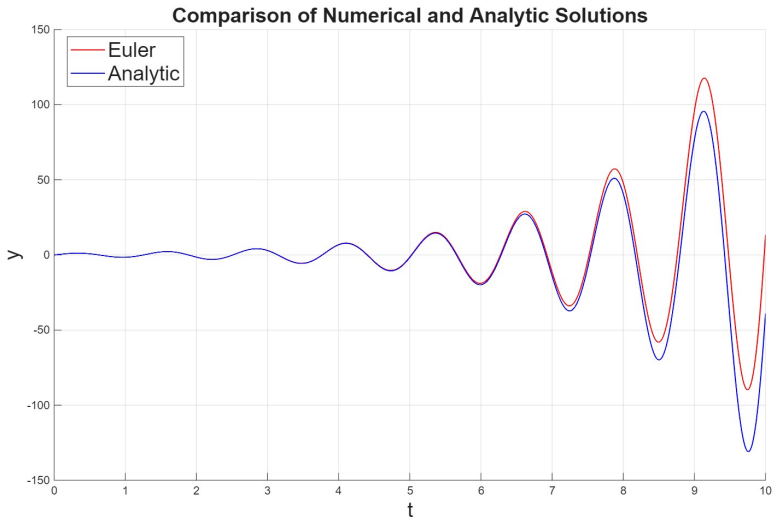

Figure 2: Euler Method, Iteration 2

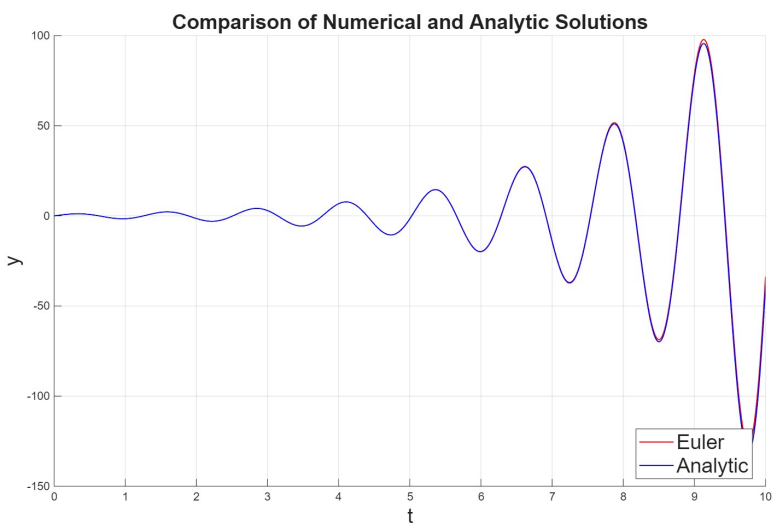

Figure 3: Euler Method, Iteration 3

As we can see the finer you make the little intervals between calculations, the nearer you get to the analytical solution.

3 1-D Heat Equation Matlab

In this section we are going to discuss a very simplified version of a heat distribution in a metal rod of length N with respect to time. As input data we become the temperature $T(t_0) = T_0$ in every point in the rod. We make the following assumptions:

- Depending on the distribution, but the start and end value are fixed and do not change.
- The PDE, being solved is: $\frac{\partial T^2}{\partial^2 x} = const * \frac{\partial T}{\partial t}$.
- The *const* in the previous point is decided by me to be 2.

In this folder I have included as well a Matlab script, which generates data with different distributions for the rod. The first example is the Gaussian distribution, then the sine function. The 3-D plot represents the $T(x, t)$.

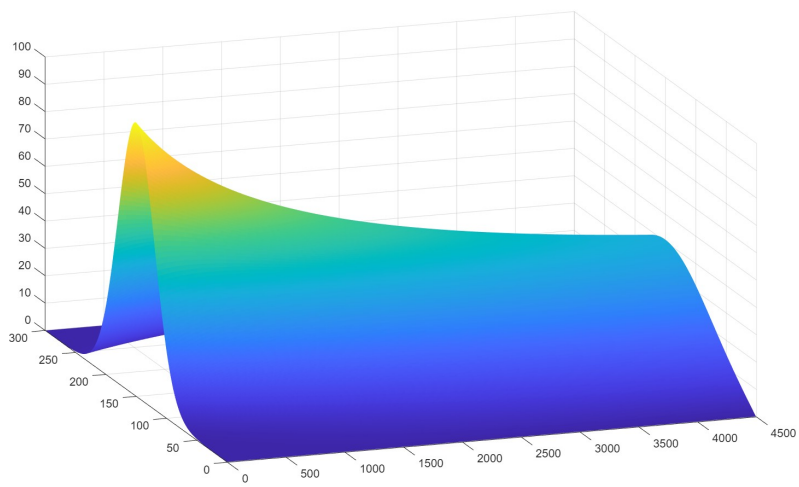

Figure 4: The Gauss distribution

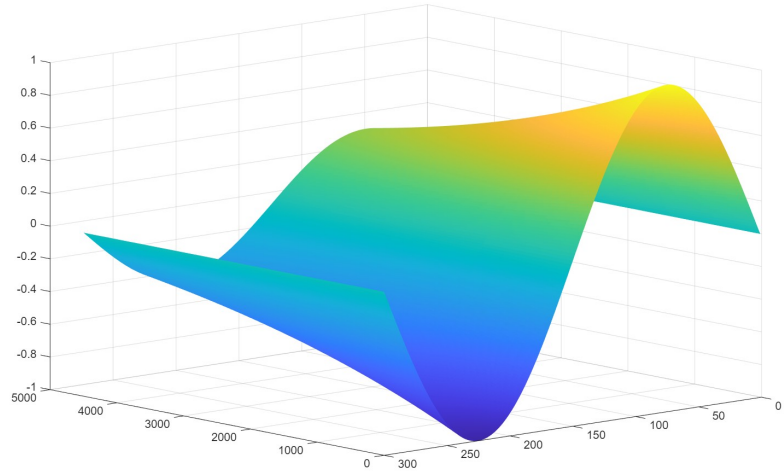

Figure 5: The $Sin()$ Function